

- 1 (a) Fig 1.1 shows the protein structure of the enzyme cellulase, which is found in animals feeding on grass such that they can digest the cellulose found in grass.



Fig. 1.1

- (i) Name structure A and describe how it is formed.

1. β -pleated sheet;

2. is made up of different sections of the polypeptide chain that run alongside each other/ are antiparallel;

3. Adjacent strands are held together by hydrogen bonds formed between $-\text{CO}$ and $-\text{NH}$ groups (of the polypeptide backbone) to form the sheet; @1m [3]

- (ii) State two differences between the structures of cellulase and its substrate.

1. Made up of amino acids vs β -glucose;

2. Peptide bond vs (β 1-4) glycosidic bond between monomers;

3. Ionic bonds, hydrophobic interactions and disulphide bonds(any 2 of 3) found in the structure of cellulase but not its substrate;

4. No rotation of amino acids vs alternate β -glucose rotated 180° ;

5. No crosslinks vs crosslinks between cellulose;

6. Globular vs fibrous;

@ 1m, max 2 [2]

(iii) Briefly describe the mode of action of cellulase.

1. Enzyme lowers activation energy;
2. Ref to acting as orientation surfaces/ ref to strain effect/ proximity effect/ microenvironment effect/ acid-base catalysis;
3. The (3D conformation of) substrate (e.g., cellulose) is complementary to the active site of the enzyme, thus forming enzyme-substrate complexes;
4. With ref. to change in 3D conformation of enzyme for a better fit/ tighter binding between substrate and enzyme;
5. Resulting in cleaving of glycosidic bond between β -glucose; [4]
@ 1m, max 4

(b) Collagen is a protein found extensively in connective tissue, bones, cartilage, dermis (skin), blood vessels and teeth. It is a fibrous protein, and has a very distinct structure and function from cellulase.

Describe **two** differences between the primary structures of collagen and cellulase.

1. Amino acid sequence in collagen is highly regular with repeating tripeptide sequence while amino acid sequence in cellulase rarely shows regularity;
2. Amino acid sequence in collagen may vary between two samples of collagen; while amino acid sequence in cellulase is highly specific;
3. Length of polypeptide chain varies in collagen while length of polypeptide chain seldom varies in cellulase;
4. More amino acids make up collagen while less amino acids make up cellulase;
5. Modified amino acids such as hydroxylysine and hydroxyproline are found in collagen but not cellulase;

@ 1 m per comparison, max 2 [2]

[Total: 11]

- 2 Matrix metalloproteinase-13 (*MMP-13*) gene is found on chromosome 11 in humans. It codes for *MMP-13*, an enzyme that is involved in the breakdown of extracellular matrix proteins such as collagen, in normal physiological processes. During inflammation, a protein, AP1, is activated and binds to a conserved DNA sequence about 20kb (kilobases) upstream of the *MMP-13* gene (Fig. 2.1) to increase (up-regulate) its expression. Histone modification is found to occur at the stretch of DNA from the conserved DNA region to the transcribed region of the *MMP-13* gene. The AP1-bound conserved DNA region is also found to come into close proximity to the *MMP-13* transcription start site.

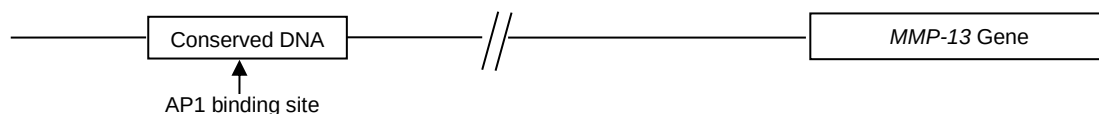


Fig. 2.1

- (a) A gene is a region along a DNA molecule that codes for a specific sequence of amino acids in a polypeptide chain. Describe the structure of DNA.
1. **Polymer of deoxyribonucleotides/composed of 4 types of deoxyribonucleotide containing base thymine, cytosine, adenine and guanine;**
 2. **two antiparallel polynucleotide strands/double helix;**
 3. **ref to complementary base-pairing by hydrogen bonds / 3 H bonds bet G-C and 2 H bonds bet A-T/purine with pyrimidine/ref to 1:1 ratio;**
 4. **width betw sugar-phosphate backbone constant and equals to the width of a purine and a pyrimidine/2 nm wide;**
 5. **One complete turn has 10 base pairs with length of 3.4 nm;**
 6. **Phosphodiester bonds between adjacent deoxyribonucleotides;** [3]
- Any 3 @ 1m each
- (b) Describe the role of the AP-1 protein in increasing the expression of the *MMP-13* gene.
1. **AP1 act as the activator (R! repressor) and binds to the conserved DNA region, the enhancer (R! silencer);**
 2. **DNA bends to bring the AP1 protein and conserved DNA region to close proximity to the promoter/transcription initiation complex/*MMP-13* gene;**
 3. **Ref. to mediator proteins and transcription factors binding at promoter/AP-1 protein;**
 4. **Increases the affinity of RNA polymerase to the promoter of the *MMP13* gene/ enhances formation of the transcription initiation complex; @ 1 m** [3]
- (c) Describe **one** type of histone modification that can occur in the presence of the activated AP1 protein.
1. ***Acetylation of histones;**
 2. **neutralises the positively-charged R groups on histones / decreases the affinity of the histone tails for DNA;**
- OR
4. ***De-methylation of histones;**
 5. **proteins that bind to methylated histones removed;**
- @ 1 m [2]
- (d) *MMP-13* is secreted as an inactive form. It is activated once the pro-domain is cleaved, leaving an active enzyme that participates in collagen degradation. State this level at which the expression of *MMP-13* gene is controlled.
1. **post-translational;** [1]
- (e) *MMP-13* is involved in the degradation of the basement membrane where cells are anchored. Tumour cells have elevated *MMP-13* production and secretion. Suggest how the elevated secretion of *MMP-13* contributes to the malignant properties of the tumour cells.
1. **Metastasis/metastasizing;**
 2. **Tumour cells freed/no longer bound/not anchored;**
 3. **Released into bloodstream;**
 4. **Resulting in formation of other/secondary tumours;** @ 1 m, max 3 [3]
- [Total: 12]

3. Fig. 3.1 below is a diagram of a dividing animal cell.

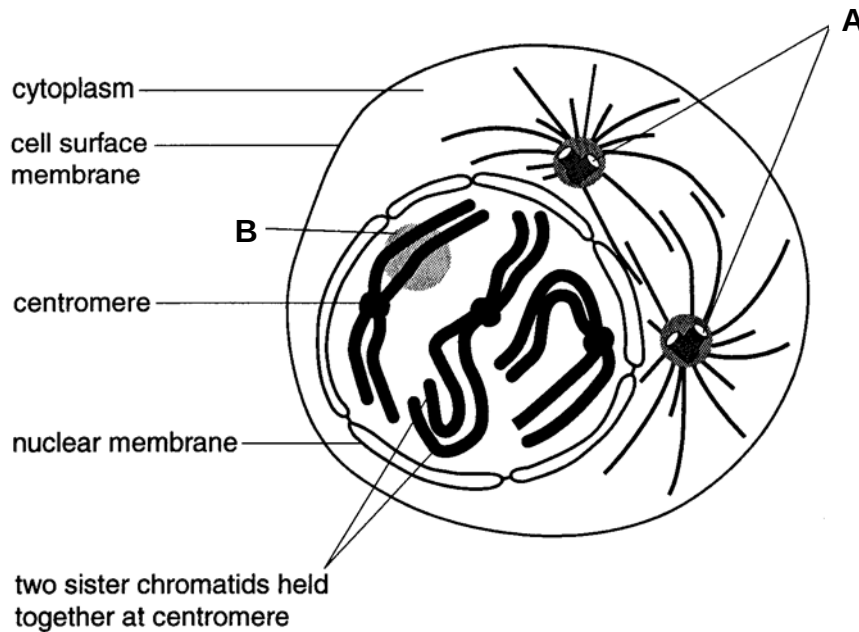


Fig 3.1

(a) Name the structures labelled **A** and **B**.

A: Centrioles; R: Asters

B: Nucleolus; @1/2 m [1]

(b) State **one** visible feature in Fig 3.1 that enables you to identify that the cell is at

(i) Prophase of (ii) Meiosis II.

(i) Presence of nuclear envelope/separating centrioles/absence of spindle/formation of spindle fibres/

Condensed chromosomes but not aligned at equatorial plane; any 1

(ii) Absence of homologous pairs of chromosomes/bivalents/1 of each type of chromosome/presence of odd number of chromosomes; @1m [2]

(c) Discuss the significance of meiosis II.

1. Separation of sister chromatids of each chromosome (into 2 daughter nuclei);

2. Results in haploid cells/each with only one of each type of chromosomes/halving the number of chromosomes;

3. Diploid number of organism will be restored/prevents doubling of chromosomes upon fertilisation;

4. Daughter cells are genetically different (due to crossing over and independent Segregation); @1 m [3]

- (d) Describe the event(s) that could possibly take place following prophase of Meiosis II, resulting in the formation of a zygotic cell with diploid number of 7 upon fertilisation.

1. * Non-disjunction;
2. One pair of sister chromatids fail to separate at anaphase II;
3. Due to failure of attachment to spindle fiber/ of centromere dividing;
4. One daughter cell with (n+1/4) and the other (n-1/2) chromosomes/OWTTE;
5. Fusion of (n+1/4) gamete with normal gamete (3) to form $2n = 7$ zygote; [3]
@ 1 m, * point essential to get full mark

[Total: 9]

- 4 Fig. 4.1 shows the components of the *lac* operon and its regulatory gene which is some distance away.

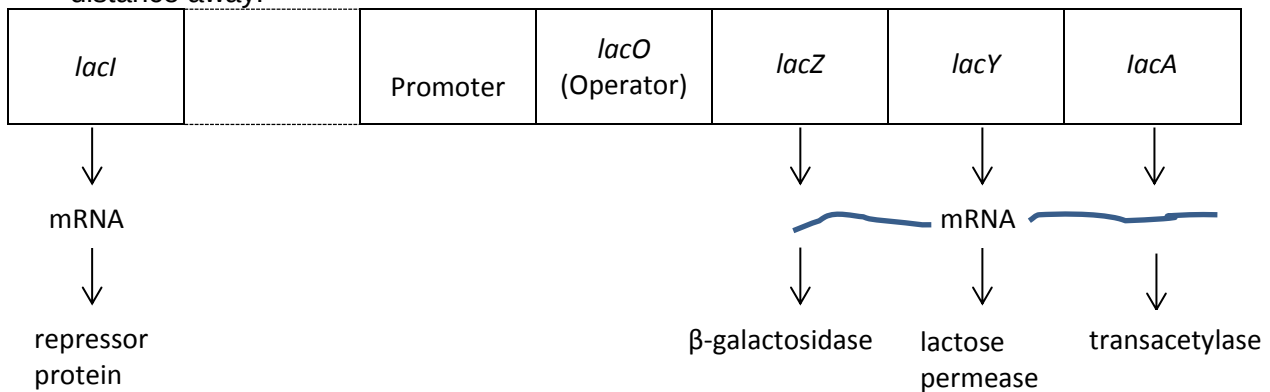


Fig. 4.1

- (a)(i) Describe what will happen to the repressor protein if lactose is present.

1. Allolactose/lactose binds to (allosteric site of) repressor protein and
2. Alters configuration/3D conformation/tertiary structure of repressor

3. repressor detaches / cannot bind to operator @1m [2]

- (ii) Explain how a bacterial cell with a diploid genotype $\frac{lacI^+ lacO^- lacZ^+ lacY^- lacA^+}{lacI^- lacO^- lacZ^+ lacY^- lacA^+}$ will respond in the presence of lactose (+: wild type allele; -: loss of function allele)

1. ($lacI^+$ is dominant over $lacI^-$) hence repressor protein is produced;

2. both copies of operator are defective;

3. (Normal) repressor cannot bind to operator

4. structural genes will be constitutively/continuously expressed

5. LacY gene for lactose permease not expressed/ lack of lactose permease/non- functional lactose permease

6. Functional β -galactosidase and transacetylase formed;

7. lactose not transported into cell (and metabolised); @1m [5]

- (b) 5'-capping and 3'-polyadenylation are two kinds of post-transcriptional modifications that are present in eukaryotes but absent in prokaryotes. Describe another post-transcriptional modification necessary to produce mature mRNA in eukaryotes.

1. Spliceosome recognises/binds splice site;
2. Bring about the removal of introns/ spliced exons together to form mature mRNA @1m

[2]

- (c) Human DNA has been analysed and details of certain genes are shown in Table 4.1.

Table 4.1

Gene	Gene size / kb	mRNA size / kb	Number of introns
Insulin	1.7	0.4	2
Albumin	25.0	2.1	14
Phenylalanine hydroxylase	90.0	2.4	12
Dystrophin	2000.0	17.0	50

- (i) Calculate the average size of the introns for the albumin gene (show your workings.)

$$\frac{25.0 - 2.1}{14} = 1.6 \text{ kb};$$

[1]

- (ii) With reference to Table 4.1, describe the relationship(s) between the gene size and the number of introns.

1. Generally bigger the gene size, the larger number of introns
2. Correct ref to data comparison of gene size and number of introns;
3. Ref to exceptions in trend as shown by phenylalanine hydroxylase / albumin gene @1m

[3]

[Total: 13]

- 5 Coffin–Lowry Syndrome (CLS) is a sex-linked dominant genetic disease that causes severe mental disability in humans. This disorder is caused by mutations in gene **R**. The normal protein coded for by this gene is required for the formation of long-term memory, and the survival of nerve cells. Lack of this protein hence affects mental development. Patients with CLS experience brief episodes of collapse when they are excited or startled by a loud noise.

- (a) A phenotypically normal woman married a CLS sufferer. Using a genetic diagram, illustrate this cross and determine the probability of the couple having a male child suffering from CLS. [3]

Penalise once for symbol other than X^R and X^r used;

Parental phenotype:	Normal female	x	CLS male
Parental genotype:	X^rX^r	x	X^RY
Gametes	X^r		X^R Y
F1 genotype:	X^RX^r		X^rY
F1 phenotypic ratio:	1 CLS female	:	1 normal male
Probability of a male child suffering from CLS:	0		

1. 1m for both genotype and gametes correct

2. 1m for both F1 genotype and F1 phenotypic ratio correct

3. 1m for correct answer

The Canine Genome Sequencing Project was successfully completed in year 2005. During the project, researchers needed to investigate two particular genes in dogs. The genes were the amino-methyl-transferase (*AMT*) gene (which codes for AMT enzyme to degrade excess glycine), and the T-cell leukemia translocation-associated (*TCTA*) gene (which codes for TCTA protein involved in osteoclastogenesis, the formation of osteoclasts for the breakdown of bone tissue).

- (b) To increase sample size for statistical reliability, scientists selected several heterozygous female poodles for normal AMT phenotype and normal TCTA phenotype, and conducted test crosses on them. The following results were observed in the offspring.

Normal AMT, normal TCTA dogs	96
Normal AMT, mutant TCTA dogs	18
Mutant AMT, normal TCTA dogs	20
Mutant AMT, mutant TCTA dogs	94

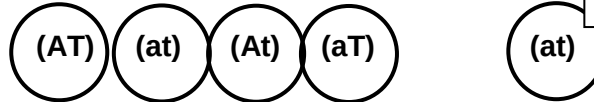
- (i) Using **A** and **a** to represent the alleles for the *AMT* gene, and **T** and **t** to represent the alleles for the *TCTA* gene, illustrate the above cross with the use of a genetic diagram. [4]

R: Linkage not indicated by brackets

Parental phenotype: normal *AMT*, normal *TCTA* x mutant *AMT*, mutant *TCTA*

Parental genotype: (AT) (at) x (at) (at)

Gametes



1. 1m for P genotype, with brackets indicating linked genes;

2. 1m for circled gametes;

F1 genotype:

(AT) (at) (at) (at) (At) (at) (aT) (at)

F1 phenotype:

Normal *AMT*,
normal *TCTA*

Mutant *AMT*,
mutant *TCTA*

Normal *AMT*,
mutant *TCTA*

Mutant *AMT*,
normal *TCTA*

3. 1m for F1 genotype;

4. 1m for corresponding phenotype;

- (ii) Calculate the recombinant frequency between the *AMT* gene and *TCTA* gene. [1]

$$\text{C.O.V.} = [20+18] / 228 \times 100 = 16.67\%$$

R: fraction

@1 mark

- (iii) The recombinant frequency between the *AMT* gene and the canine *DAG1* gene was found to be 8.4%, and the recombinant frequency between the *TCTA* gene and *DAG1* gene was found to be 8.3%. On the diagram of a chromosome below, indicate the relative positions of the *AMT*, *TCTA* and *DAG1* genes. [1]



Ans: *AMT* gene, *DAG1* gene, *TCTA* gene

OR *TCTA* gene, *DAG1* gene, *AMT* gene

[Total: 9 marks]

- 6 Other than activating adenylyl cyclase, activated G-protein-coupled-receptors (GPCRs) can activate another signalling pathway. Fig. 6.1 below shows how the activation of GPCR eventually leads to the activation of protein kinase C.

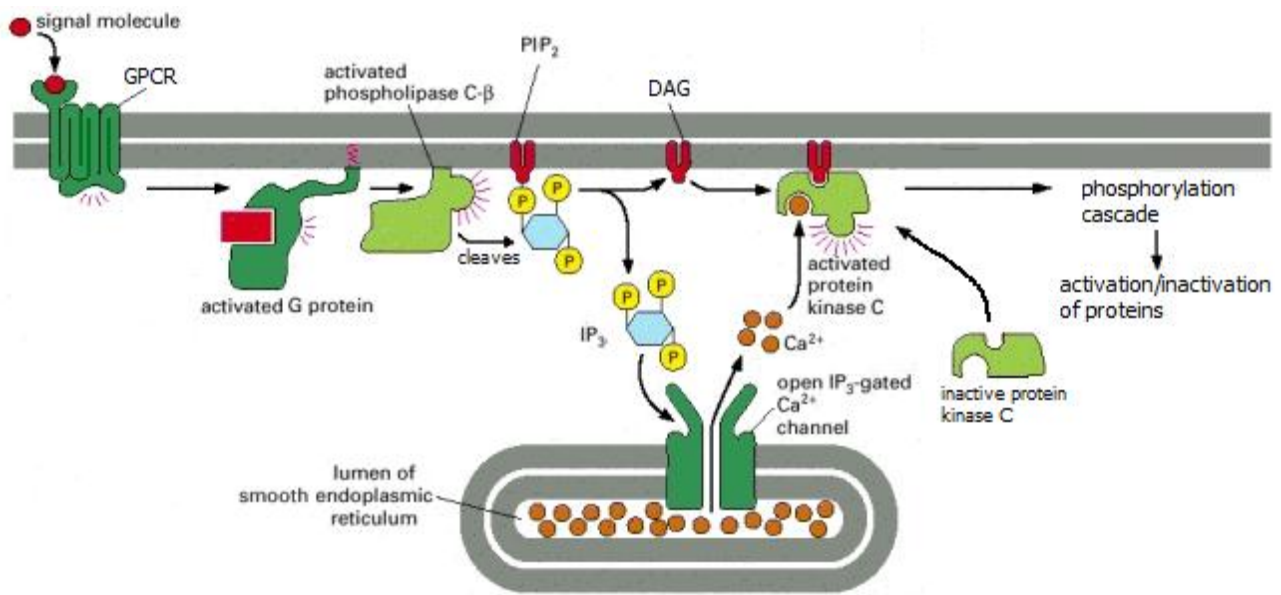


Fig. 6.1

- (a) Describe how G protein is activated upon binding of the signal molecule to GPCR.

1. GPCR changes conformation/3D shape;

2. Binds to G protein, and causes the displacement of GDP with GTP;

@ 1m

[2]

- (b) With reference to Fig. 6.1, describe how the activated phospholipase C-β results in the activation of protein kinase C.

1. Activated phospholipase C-β cleaves PIP₂ into IP₃ and DAG;

2. IP₃ binds causes the IP₃-gated Ca²⁺ channel to open, allowing efflux of Ca²⁺ from SER (into cytoplasm)/ increasing [Ca²⁺] in the cytoplasm;

3. Ca²⁺ and DAG bind to protein kinase C (and activate it);

@ 1m

[3]

- (c) Explain what is meant by the term 'phosphorylation cascade'.

1. Phosphorylation cascade refers to a series of protein kinases which phosphorylate other proteins/ in sequence/series;

2. Ref. to signal amplification;

@ 1m

[2]

(d) Other than through the phosphorylation cascade, explain how the signal is amplified in the pathway shown in Fig. 6.1.

1. **amplification occurs when (one) signal molecule/activated GPCR activates many G proteins;**
2. **when (one) activated phospholipase C- β converts many molecules of PIP₂ into IP₃ and DAG;**
3. **and when (one) IP₃ molecule results in many Ca²⁺ ions entering the cytoplasm/activation of many protein kinase C;** @ 1m [2]

(e) Besides storing Ca²⁺, describe **two** other functions of the smooth endoplasmic reticulum.

1. **synthesis of lipids/steroids/triglycerides/phospholipids;**
2. **detoxification of drugs/poisons (in liver cells);**
3. **carbohydrate metabolism/breakdown of glycogen (in liver cells); any 2 @ 1m** [2]

Some types of protein kinase C do not require presence of Ca²⁺ for activation. Protein kinase C is known to be involved in cell signalling pathways which promote cell division.

(f) Phorbol esters are plant compounds which mimic the action of DAG. Phorbol esters are known to promote the formation of tumours. With reference to Fig. 6.1, suggest and explain how phorbol esters can promote tumour formation.

1. **Phorbol esters bind to/activate protein kinase C;**
2. **(activated) protein kinase C activates proteins/genes/cell signal pathways which promote cell division / prevent apoptosis;** @ 1m [2]

[Total: 13]

7 The mechanisms of speciation in ferns have been studied in temperate and tropical habitats. One group of three species from the genus *Polypodium* lives in rocky areas in temperate forests in North America. Members of this group have similar morphology (form and structure). Another group of four species from the genus *Pleopeltis* live at different altitudes in tropical mountains in Mexico and Central America. Members of this group are morphologically distinct.

Data from the different species within each group was compared in order to study the mechanisms of speciation.

Genetic identity was determined by comparing the similarities of certain proteins and genes in each species. Values between 0 and 1 were assigned to pairs of species to indicate the degree of similarity in genetic identity. A value of 1 would mean that all the genetic factors studied were identical between the species being compared.

Fig. 7.1 shows the approximate distribution in North America of the first group comprising three species of *Polypodium* (*Po.*) and a summary of genetic identity.

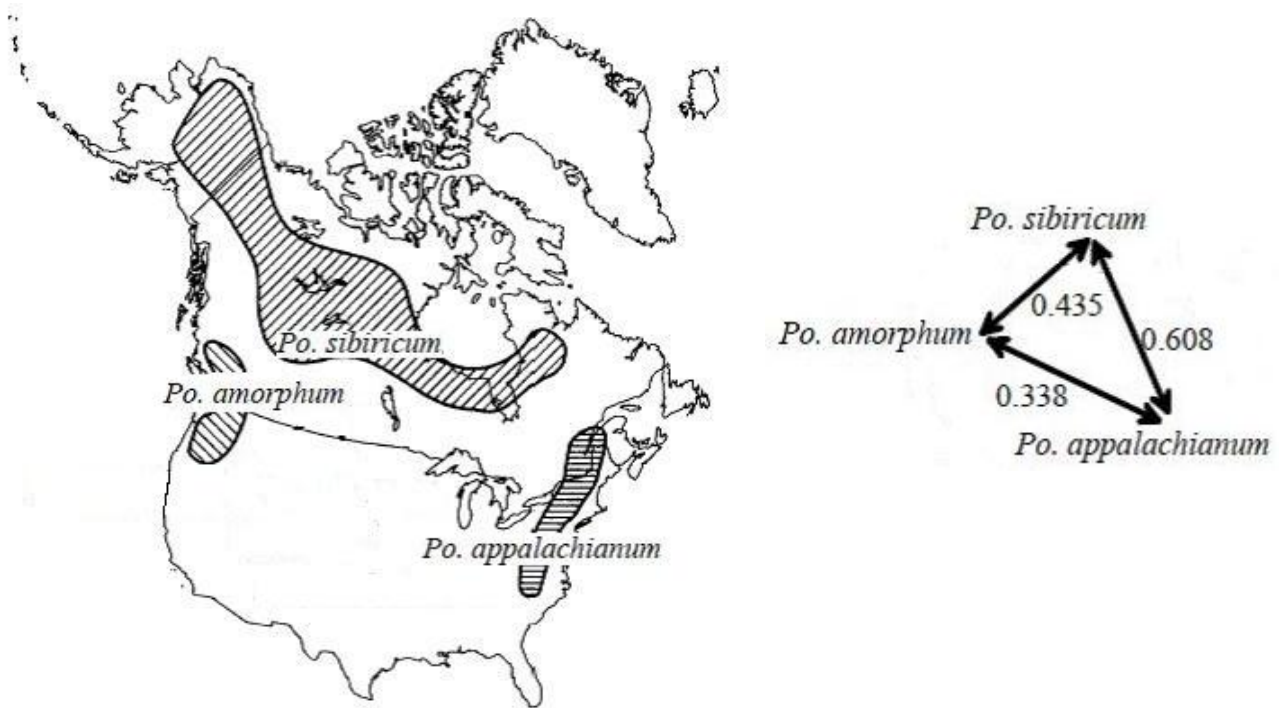


Fig. 7.1

Fig. 7.2 shows the approximate distribution in Central America and Mexico of the second group comprising four species of *Pleopeltis* (Pl.) and a summary of genetic identity.

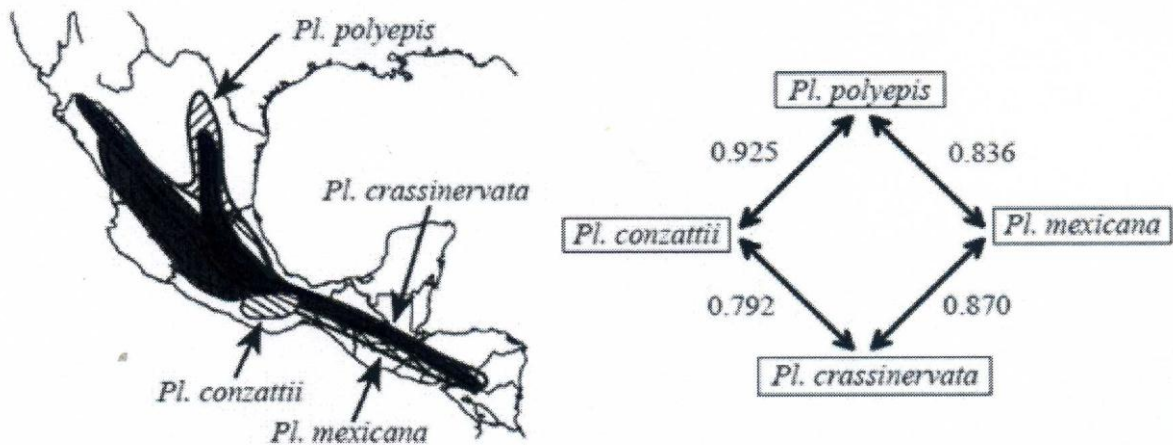


Fig. 7.2

(a) State, with reasons which group, *Polypodium* or *Pleopeltis*, has most probably been genetically isolated for a longer period of time.

1. *Polypodium*;

2. it has lower genetic identity values of 0.338 – 0.608/the highest genetic value of 0.608 in *Polypodium* is still lower than those of *Pleopeltis*/ the lowest genetic value of *Pleopeltis* is still higher than those of *Polypodium*;

3. takes time to accumulate mutations / genetic changes;

4. Isolation due to geographical barrier limiting gene flow;

@ 1m [4]

(b) Describe the distributions of the two groups of ferns within their respective geographical locations.

1. The *Polypodium* species are (completely) found in different parts of the continent/geographically separated;

2. The *Pleopeltis* species are much closer together / physically overlapping / share same habitats;

@ 1m [2]

(c) Suggest how the process of speciation could have occurred in *Polypodium*.

1. Allopatric speciation;

2. The three species of *Polypodium* are geographically isolated (on different parts of the continent);

3. Inherited variation is present in the (original) population of *Polypodium*;

4. Different habitats exert different selection pressures;

5. Only those with favourable traits are selected for and survive to pass on their alleles to their offspring;

6. No interbreeding/gene flow between the different populations;

7. Separation of gene pools of isolated populations/change in allele frequencies over time;

@ 1m, max 5 [5]

(d) According to neutralists, when one compares the genomes of existing species, the vast majority of molecular differences, which arises from gene mutations, are selectively neutral. Describe the effect of changing the selection pressure on populations of species which have these neutral mutations.

1. These mutations do not affect fitness/reproductive success;

2. These organisms do not come under selection pressure/changing selection pressure has no effect;

3. Neutral mutations would either become fixed in all of its members, or be lost entirely due to genetic drift.

@ 1m [2]

[Total: 13]

Section B

- 8 (a)** Describe the roles of organelles involved in the synthesis and transport of proteins destined to be embedded on the cell surface membrane. [8]

1. Ref to gene coding for membrane protein transcribed in nucleus;
2. Ref. to export of mRNA to cytoplasm via nuclear pores;
3. mRNA attaches to ribosome at RER surface for translation;
4. Ref to formation of translation initiation complex involving methionyl-tRNA, ribosomes at the 5'-end of mRNA;
5. Ref to complementary base pairing between anti-codon of tRNA with codons of mRNA;
6. Ref to (elongation via) peptidyl transferase catalyzing the formation of peptide bond between amino acids;
7. As ribosome moves along mRNA in 5' to 3' direction till stop codon;
8. polyribosomes may form to increase the rate of protein synthesis;
9. Completed polypeptide released into and embedded into membrane of RER;
10. Transport vesicles with polypeptide (embedded) moved towards and fused with (cis face of) Golgi apparatus;
11. Proteins undergo post-translational modifications in RER/GA;
12. Membrane (with embedded proteins) pinches off as secretory vesicles;
13. Vesicles travel along microtubules;
14. And fuse with plasma membrane/ cell surface membrane;

@ 1m , max 8

- (b)** Compare the process in which ATP molecules are synthesised during respiration with that in photosynthesis. [6]

Similarities:

1. Both processes involved chemiosmotic synthesis/proton gradient across membrane surfaces;
2. Both processes involved hydrogen carriers, ETC and ATP synthase (any 2 of 3);

Differences:

Respiration	Photosynthesis
3. ATP synthesized via both substrate-level and oxidative phosphorylation;	ATP synthesized via photophosphorylation only;
4. ATP produced in cytoplasm as well as inside mitochondria;	ATP produced only in chloroplasts;
5. Energy/(H ⁺ and) electrons used for ATP synthesis from chemical reactions/oxidation involving organic compounds;	Energy/H ⁺ and electrons used for ATP synthesis from light absorbed by photosynthetic pigments/photolysis of water;
6. Proton pool maintained in intermembrane space in mitochondria / Ref to direction of proton movement/location of ATP synthase/ETC;	Proton pool maintained in thylakoid space in chloroplasts;
7. ATP formation during aerobic respiration leads to the formation of water using O ₂ as final electron acceptor;	ATP formation during photosynthesis leads to the formation of reduced NADP and release of O ₂ during non-cyclic photophosphorylation;
8. while lactic acid (in mammalian cells) and ethanol + CO ₂ (in plant and yeasts) formed during anaerobic respiration;	

9. ATP produced continuously/both in presence or absence of light/in all cells;	ATP production only in the presence of light/cells with chloroplasts;
10. Hydrogen carriers involved are NAD ⁺ and FAD;	Hydrogen carriers involved are NADP ⁺ ;
11. Process results in oxidation of reduced NAD ⁺ and FAD;	Process results in reduction of NADP ⁺ ;

@ 1 m max 5

(c) Describe how a phylogenetic tree is constructed.

[6]

1. Involves a comparison of nucleotide sequences / amino acid sequences of different groups of organisms;
2. Ref to the method of DNA hybridization to compare nucleotide differences;
3. Genes / proteins being compared must be homologous;
4. Grouped according to degree of similarities / with highest similarity index/closely related species are grouped/ clustered together;
5. Branch length represents the number / degree of changes that have occurred;
6. Common ancestor between species is represented by the nodes in the phylogenetic tree;
7. A common ancestor together with all its descendants are organized into a clade
8. Ref to use of molecular clock to elucidate time of divergence;
9. Ref to use of radiometric dating/stratigraphy/index fossils to find age of fossils;
10. Ref. to use of transition fossils to verify/support phylogeny;

@1m, max 6

9 (a) With reference to the control of blood glucose concentration, explain the principles of homeostasis in terms of receptors, effectors, and negative feedback. [8]

- 1 Homeostasis is the maintenance of a constant internal environment;
- 2 Blood glucose concentration is maintained at 80-90 mg/100 ml blood;

Receptor

- 3 β/α cells of Islets of Langerhans are the receptors (which detect the increase/decrease/deviation in blood glucose concentration);
- 4 Resulting in the (increased) secretion of insulin/glucagon;

Effector

- 5 The effectors are liver, fat and muscle (any 2 of 3 for insulin) / liver (for glucagon);
- 6 Any e.g. of corrective mechanism by effector i.e. insulin causes increase in glycogenesis in liver/muscle cells / increase in glucose carriers in the plasma membrane / increase in glucose uptake in fat/muscle cells / increase in rate of respiration/ATP synthesis/ DNA/RNA synthesis OR glucagon causes increase in glycogenolysis/glucoseoneogenesis / decrease in glycogenolysis in liver cells ;
- 7 Insulin/glucagon brings about a decrease/increase in blood glucose levels (back to norm);

Negative feedback

- 8 Ref. to negative feedback e.g. a disturbance in the system sets in motion a sequence of events which counteracts the disturbance and brings the system to its original state/ Negative feedback brings about change in the opposite direction restoring stability / Negative feedback is seen when the initial increase in blood glucose concentration brings about the subsequent decrease in blood glucose concentration / OWTTE;

- 9 (Negative feedback mechanism is also involved in inhibiting) further secretion of insulin/glucagon as receptor/ α/β cells of Islets of Langerhans detects that blood glucose concentration is back to norm;
- 10 Ref. made to the opposite scenario when blood glucose concentration decreases/increases below/above set point, resulting in the release of the other hormone (glucagon / insulin);
- 11 Ref. to self-regulating nature of homeostasis;

Note: students need to have at 1 point from 3-5 (receptor section) AND 5-6 (effector section) AND 8-9 (negative feedback section) to score full marks.

@1m, max 8

- (b) Describe how the uni-directional transmission of nerve impulses is ensured along a neuron and across a synapse. [6]

Along a neuron (max 3)

1. Ref to AP/repolarisation phase, prolonged efflux of K⁺ results in hyperpolarisation;
2. Leading to refractory period;
3. where the membrane cannot respond to another normal stimulus/be depolarised;
4. Absolute refractory period (when no stimulus, however strong, can initiate another AP) and relative refractory period (when another AP can only be initiated if the stimulus is stronger than normally required);
5. Ensures that the AP is propagated towards the synaptic terminal;

Across a synapse (max 3)

6. Synaptic vesicles/neurotransmitter present only in pre-synaptic terminals;
7. Neurotransmitter receptors / ligand-gated ion channels are present only on the post-synaptic membrane / dendrites / cell body of the post-synaptic neuron;
8. Acetylcholinesterase present in cleft/at post-synaptic membrane to hydrolyse neurotransmitter to maintain diffusion gradient;
9. Hence impulse/neurotransmitter is transmitted only from pre-synaptic terminal to post-synaptic neuron;

@ 1m

- (c) With reference to sickle-cell anemia, explain how gene mutation may affect the phenotype of a person. [6]

1. The mutation in sickle-cell anemia involves the substitution of one base, from thymine to adenine in haemoglobin;
2. (Alteration in nucleotide sequence) changes the codon of mRNA (from GAA to GUA and thus the amino acid sequence in a polypeptide);
3. Mutated beta-globin/haemoglobin has hydrophilic amino acid glutamic acid replaced by hydrophobic valine;
4. Affects the conformation/tertiary structure of protein (may result in loss of protein activity/change in function/ non-functional protein, leading to a change in phenotype of a person);
5. At low oxygen concentrations, (the hydrophobic portion of HbS molecules stick together, causing) the haemoglobin (HbS) molecules (to) polymerize into fibres.

(This in turn causes) the red blood cells (to) change (from a circular biconcave shape) to a sickle-shape;

6. Sickle-shaped red blood cells may obstruct blood capillaries, hence, (multiple) organs deprived of oxygen/resulting in their damage.
7. These cells (more rigid and fragile than normal red blood cells) haemolyse readily resulting in anaemia.
8. They also accumulate in spleen for destruction leading to an enlargement of the spleen.

Points 5 to 8 (phenotype) – max 3 m

@1m, max 6